

ACQUAROSSA-COMPROVASCO, Switzerland

WMO: 067560

Lat: 46.4594N

Lon: 8.9356E

Elev: 1811

StdP: 13.76

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	25.0	-0.9	5.6	32.5	3.8	7.1	34.1	21.8	43.7	19.5	43.3	3.5	340	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	84.6	67.3	81.5	66.1	78.8	64.4	70.1	79.6	68.4	77.5	66.8	75.3	8.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	106.5	74.2	65.4	100.3	72.2	63.9	95.1	70.5	35.2	79.7	33.7	77.7	32.4	75.7	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	15.5	13.1	DB	18.5	89.0	3.6	3.5	15.9	91.4	13.8	93.5	11.7	95.4	9.1	97.9	(4)
(5)			WB	15.8	72.9	3.6	2.2	13.3	74.5	11.2	75.9	9.1	77.1	6.5	78.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	36.0	38.3	45.7	51.4	57.9	64.9	67.9	66.7	59.9	52.6	43.6	37.5		
	DBStd	12.44	5.29	6.05	6.56	5.85	5.26	5.59	4.54	4.67	4.75	5.33	5.66	5.77		
	HDD50	1610	434	330	165	52	4	0	0	0	1	34	201	389		
	HDD65	5050	897	748	599	409	227	70	23	34	163	385	643	852		
	CDD50	2321	2	3	31	94	249	447	556	517	298	114	8	2		
	CDD65	286	0	0	0	0	7	67	114	87	11	0	0	0		
	CDH74	2055	0	0	0	12	88	539	820	535	57	4	0	0		
	CDH80	436	0	0	0	0	9	118	179	127	3	0	0	0		
(14) Wind	WSAvg	4.3	4.1	4.5	5.2	5.4	4.9	4.7	4.3	3.8	3.8	3.7	3.8	3.9		
(15) Precipitation	PrecAvg	61.5	2.7	2.4	2.8	5.1	6.4	6.6	5.8	6.8	6.7	6.5	6.9	2.9		
	PrecMax	85.2	8.9	9.3	9.6	15.4	19.4	15.3	12.8	11.4	18.1	24.2	28.0	8.3		
	PrecMin	32.5	0.2	0.2	0.1	0.6	1.5	1.7	1.4	1.1	0.6	0.4	0.5	0.0		
	PrecStd	13.3	2.1	2.0	2.2	3.6	3.9	3.3	2.5	2.4	5.0	4.9	6.4	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.0	60.4	69.9	75.5	80.7	87.6	88.1	89.6	79.3	72.0	62.6	58.2		
		MCWB	45.5	43.9	50.2	56.4	62.0	67.4	67.4	67.4	63.5	57.8	48.0	43.8		
	2%	DB	51.0	54.5	64.8	70.7	76.2	83.4	84.5	83.2	75.3	67.2	56.8	52.1		
		MCWB	40.1	42.0	47.8	53.0	59.4	66.2	67.9	67.5	62.9	55.1	46.4	40.4		
	5%	DB	46.6	50.5	60.9	67.0	72.8	80.1	81.6	79.5	71.9	63.6	53.6	48.1		
		MCWB	37.4	39.9	46.3	51.1	57.6	64.8	66.4	65.9	60.5	54.3	46.0	39.7		
	10%	DB	43.2	47.2	56.8	63.2	69.3	76.7	78.7	76.4	68.9	60.3	51.2	45.2		
		MCWB	35.8	38.0	44.3	49.4	56.1	63.0	65.0	64.4	58.8	53.9	45.5	38.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.3	46.6	51.5	57.4	65.0	71.8	73.0	71.8	67.1	60.8	52.5	46.6		
		MCDB	57.1	55.3	64.2	71.8	76.0	82.2	82.3	81.2	75.1	65.4	55.7	51.2		
	2%	WB	41.6	43.1	49.4	54.9	61.9	68.9	70.3	69.7	64.7	58.7	50.3	43.9		
		MCDB	47.9	51.3	59.9	66.3	71.4	78.9	79.5	78.9	71.8	62.8	53.1	48.0		
	5%	WB	39.2	41.1	47.8	52.9	59.8	66.8	68.5	67.9	62.8	57.0	48.8	41.8		
		MCDB	44.5	48.2	57.4	62.7	68.5	76.1	77.7	76.2	68.7	60.6	51.1	45.6		
	10%	WB	37.2	39.3	46.0	51.1	57.9	64.8	66.9	66.3	61.1	55.4	47.2	39.9		
		MCDB	41.5	45.3	54.7	59.2	65.8	73.6	75.7	73.6	66.2	58.7	49.6	44.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	13.6	17.1	17.6	17.9	19.3	20.0	18.4	16.6	13.9	11.8	11.3		
		MCDBR	16.6	18.6	23.3	24.9	24.9	24.4	23.9	23.0	21.0	18.5	15.8	15.3		
		MCWBR	9.4	10.2	11.3	11.3	11.5	11.5	10.9	10.9	10.7	9.8	9.0	8.6		
	5% WB	MCDBR	14.3	16.6	20.3	20.7	20.7	21.5	21.2	20.1	18.1	14.2	11.1	12.4		
		MCWBR	9.0	9.8	10.5	10.2	10.7	11.5	11.2	10.7	10.1	8.4	7.3	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.324	0.383	0.448	0.466	0.479	0.470	0.458	0.427	0.389	0.329	0.286		
	taud		2.253	2.149	2.025	1.951	2.031	2.064	2.104	2.150	2.199	2.256	2.314	2.294		
	Ebn at Noon		252	263	261	255	255	252	253	250	246	237	234	239		
	Edh at Noon		25	35	46	55	52	51	49	44	39	32	24	22		
(44) All-Sky Solar Radiation	RadAvg		510	817	1203	1418	1523	1791	1792	1553	1209	801	496	425		
	RadStd		64	96	174	228	151	157	154	136	117	95	75	55		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.49	N/A	N/A	N/A	+1.19	N/A	N/A	N/A	+139	+79
(47)	Regional (119 neighbors)	N/A	N/A	N/A	+1.31	+0.74	+0.56	N/A	N/A	+106	+29

Nomenclature: See separate page